

Exam Date & Time: 17-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS -JANUARY 2024

SUBJECT: ECE 2122- NETWORK ANALYSIS

NETWORK ANALYSIS [ECE 2122]

Marks: 50

Duration: 180 mins.

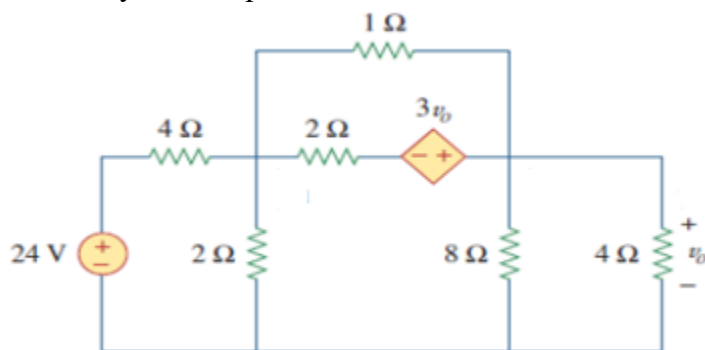
A

Answer all the questions.

Missing data may be suitably assumed. Do not use Laplace Transform unless specified.

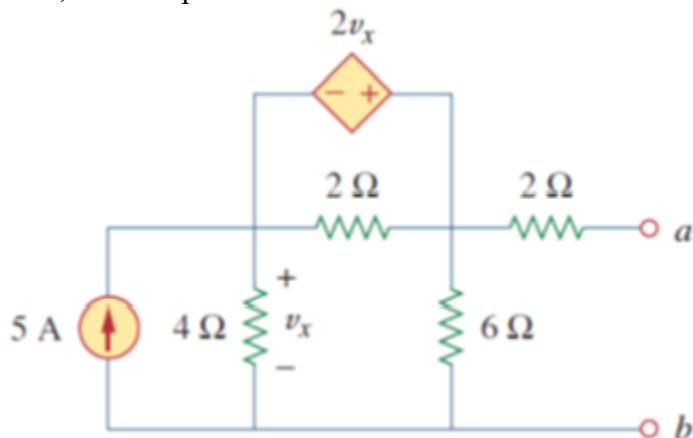
Do not use Laplace Transform unless specified.

- 1A) Determine all the mesh currents in the circuit shown in the figure. What is the power delivered by the independent source?



(4)

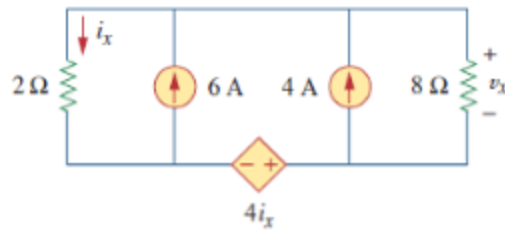
- 1B) Find the value of the load resistor connected across ab for maximum power delivery to it. Also, find the power delivered.



(4)

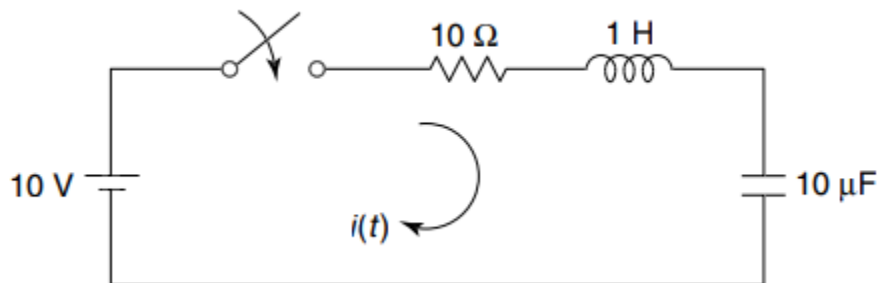
- 1C) Find v_x in the circuit shown using the superposition principle.

(2)



- 2A) In the network shown in the Figure, the switch is closed. Assuming all initial conditions as zero, find

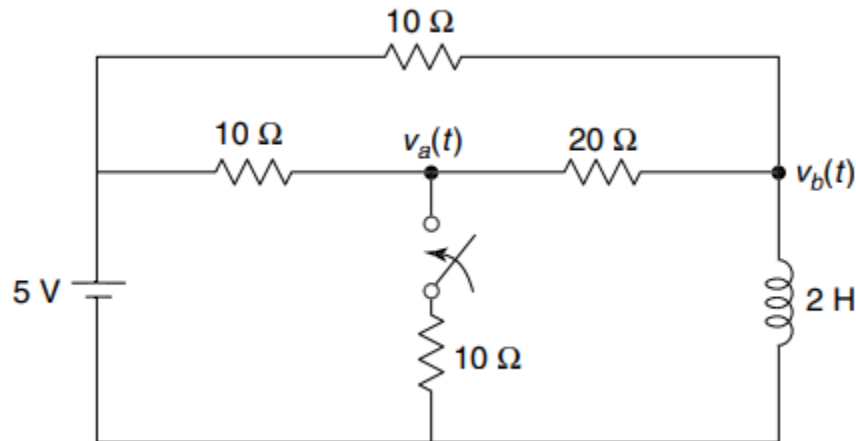
$$i, \frac{di}{dt} \text{ and } \frac{d^2i}{dt^2} \text{ at } t = 0^+.$$



(4)

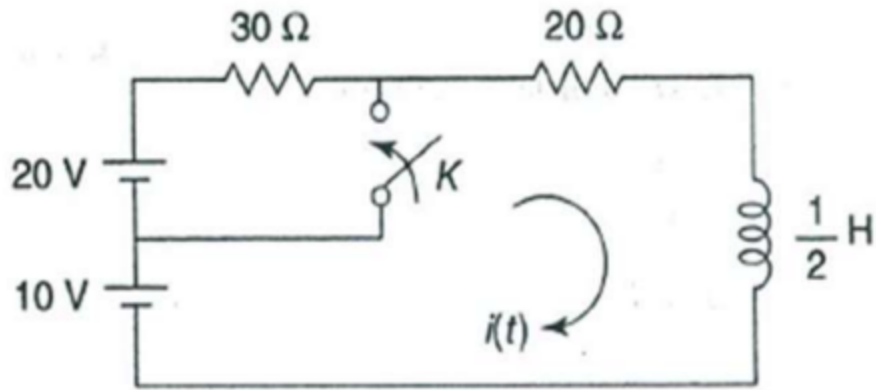
- 2B) In the network shown in the Figure, a steady state is reached with the switch open. At $t = 0$, the switch is closed. For the element values given, determine the value of

$$v_a(0^-), v_a(0^+), v_b(0^-) \text{ and } v_b(0^+)$$

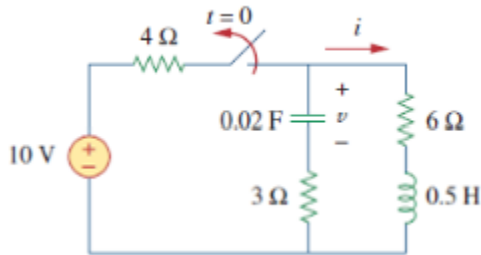


(4)

- 2C) In the network shown in the Figure, a steady state is reached with the switch K open. At $t=0$, the switch is closed. Find the current $i(t)$ for $t > 0$. (2)

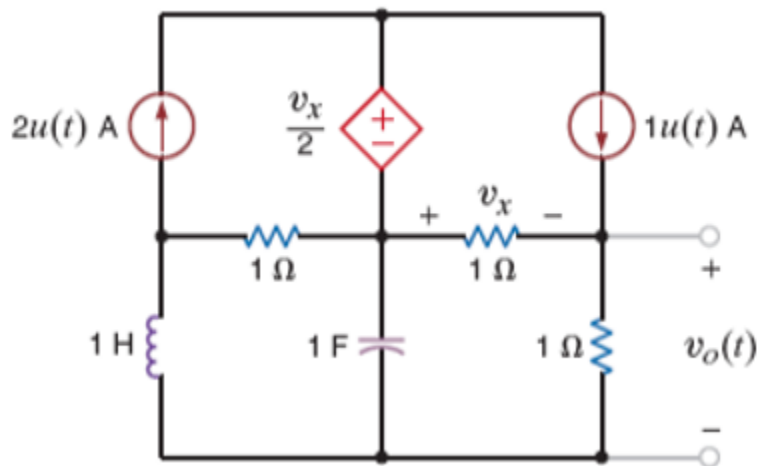


- 3A) In the circuit shown in figure below switch is opened at $t = 0$, before which steady state has been reached. Find the expression for $v(t)$ for $t \geq 0$ using Laplace transform.



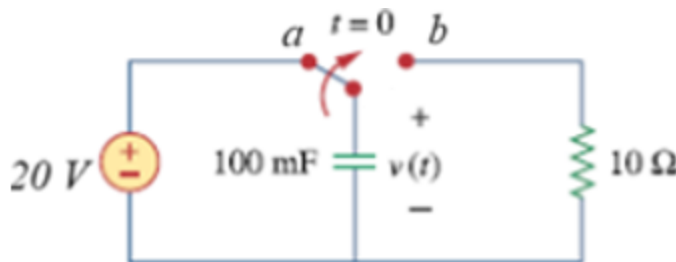
(4)

- 3B) Find $v_o(t)$ for $t > 0$, in the network in shown in figure using Laplace Transform.



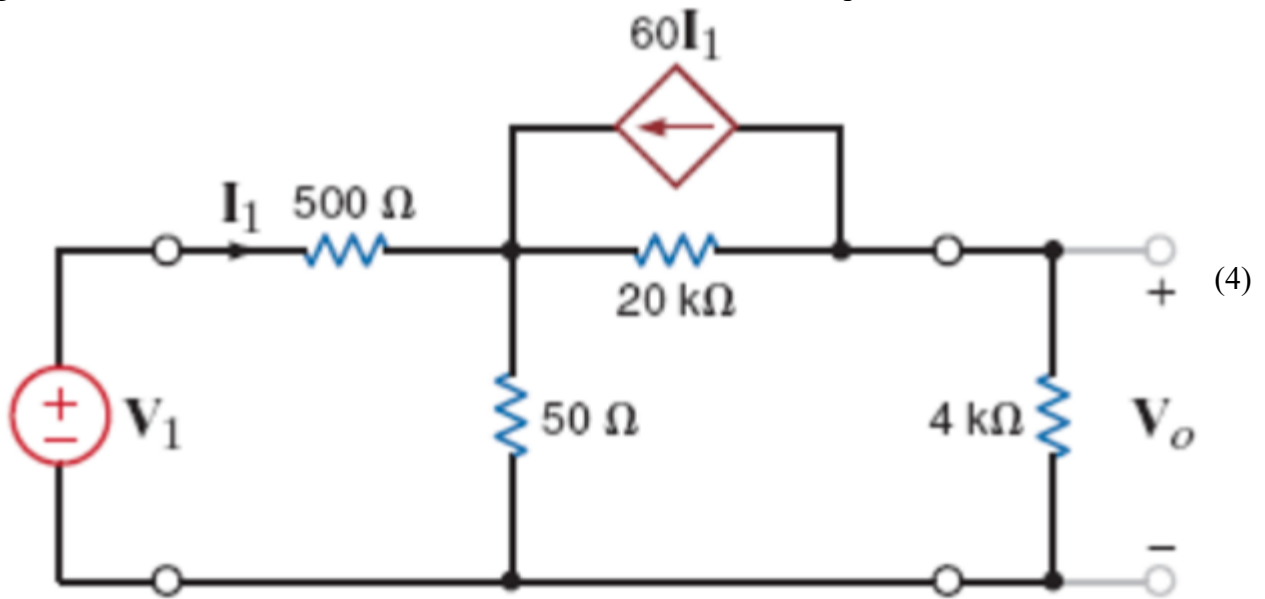
(4)

- 3C) In the circuit shown in figure below switch is moved from a to b at $t = 0$, before which steady state has been reached. Find the expression for $v(t)$ for $t \geq 0$ using Laplace transform.

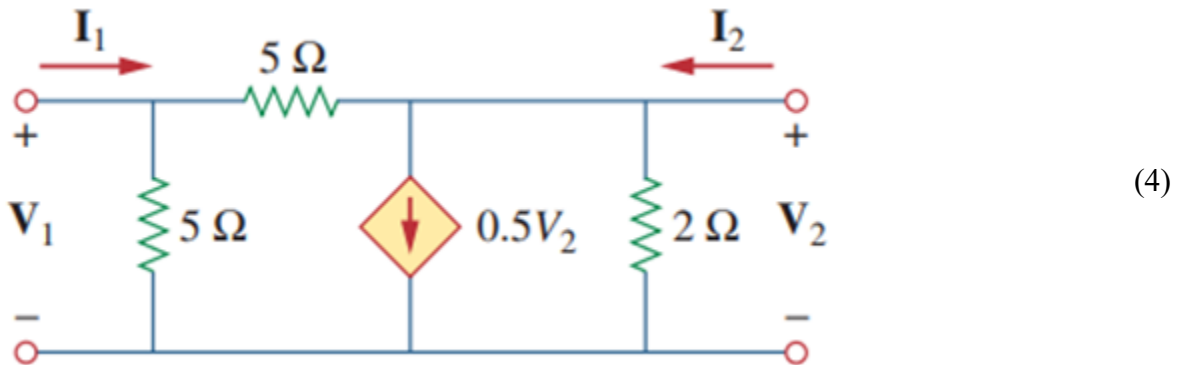


(2)

- 4A) A 20 Hz symmetrical square wave is fed to an amplifier with a peak-to-peak amplitude of 10 V. Calculate and plot the output waveform if the lower 3-dB frequency is 6 Hz. (4)
- 4B) A symmetrical square wave of amplitude $\pm 5V$ and frequency 2kHz is impressed on a low pass RC circuit. If $R=5k\Omega$ and $C=0.1\mu F$, calculate and plot the steady state output with respect to time. (4)
- 4C) A pulse of 10 V amplitude and duration 1ms is applied to a high-pass RC circuit with $R = 20k\Omega$ and $C = 0.5 \mu F$. Plot the output waveform to scale and calculate the per cent tilt in the output. (2)
- 5A) Find the Z parameters for the two-port network shown in Figure and determine the voltage gain of the entire circuit with a 4-kilo ohm load attached to the output. (4)



- 5B) Obtain the y parameters of the two-port network in Figure (4)



- 5C) The following are the Y parameters for a network. (2)

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{5}{11} & -\frac{2}{11} \\ -\frac{2}{11} & \frac{3}{11} \end{bmatrix}$$

find the Z parameters.

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